



**TRIOS CYBER**

## **LABORATORY PRACTICE 24**

Wazuh Rule and Alert Analysis

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TRIOS CYBER

### 3. Alert 2 - Integrity Checksum Changed

**Rule ID:** 550 **Alert level:** 7 **Agent:** Windows-Agent (ID 002, 10.10.20.30)  
**Description:** Integrity checksum changed. **Timestamp:** Sep 24, 2026 @ 21:10:36.667

This file-integrity alert shows that Wazuh detected a real-time modification of **C:\Users\matthew\Documents\TriosFIMLab\day23\_test.txt**. The record preserves the modification state, changed attributes, file size before and after, integrity hashes, user information, and **syscheck.mode: realtime**. The event is mapped to MITRE ATT&CK; T1565.001 (Stored Data Manipulation).

**Recommended first triage action:** Confirm whether the file modification was authorized. Review the affected path, modification time, before/after hashes, changed attributes, and the user associated with the change.

Document Details
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Table
JSON

|                   |                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index             | wazuh-alerts-4.x-2026.09.25                                                                                                                                                                                                                                                                                                                                    |
| agent.id          | 002                                                                                                                                                                                                                                                                                                                                                            |
| agent.ip          | 10.10.20.30                                                                                                                                                                                                                                                                                                                                                    |
| agent.name        | Windows-Agent                                                                                                                                                                                                                                                                                                                                                  |
| decoder.name      | syscheck_integrity_changed                                                                                                                                                                                                                                                                                                                                     |
| full_log          | File 'c:\users\matthew\documents\triosfimlab\day23_test.txt' modified<br>Mode: realtime<br>Changed attributes: size, mtime, md5, sha1, sha256<br>Size changed from '80' to '113'<br>Old modification time was: '1798302233', now it is '1798302233'<br>Old md5sum was: '9fa301d3c1535d9926aafd7189316f02'<br>New md5sum is: '4865668a965a5a0913a345cca5eacc40' |
| id                | 1798302236.316483                                                                                                                                                                                                                                                                                                                                              |
| input.type        | log                                                                                                                                                                                                                                                                                                                                                            |
| location          | syscheck                                                                                                                                                                                                                                                                                                                                                       |
| manager.name      | trioscyber                                                                                                                                                                                                                                                                                                                                                     |
| rule.description  | Integrity checksum changed.                                                                                                                                                                                                                                                                                                                                    |
| rule.firedtimes   | 1                                                                                                                                                                                                                                                                                                                                                              |
| rule.gdpr         | II.5.1.f                                                                                                                                                                                                                                                                                                                                                       |
| rule.gpg13        | 4.11                                                                                                                                                                                                                                                                                                                                                           |
| rule.groups       | ossec, syscheck, syscheck_entry_modified, syscheck_file                                                                                                                                                                                                                                                                                                        |
| rule.hipaa        | 164.312.c.1, 164.312.c.2                                                                                                                                                                                                                                                                                                                                       |
| rule.id           | 550                                                                                                                                                                                                                                                                                                                                                            |
| rule.level        | 7                                                                                                                                                                                                                                                                                                                                                              |
| rule.mail         | false                                                                                                                                                                                                                                                                                                                                                          |
| rule.mitre.id     | T1565.001                                                                                                                                                                                                                                                                                                                                                      |
| rule.mitre.tactic | Tenact                                                                                                                                                                                                                                                                                                                                                         |

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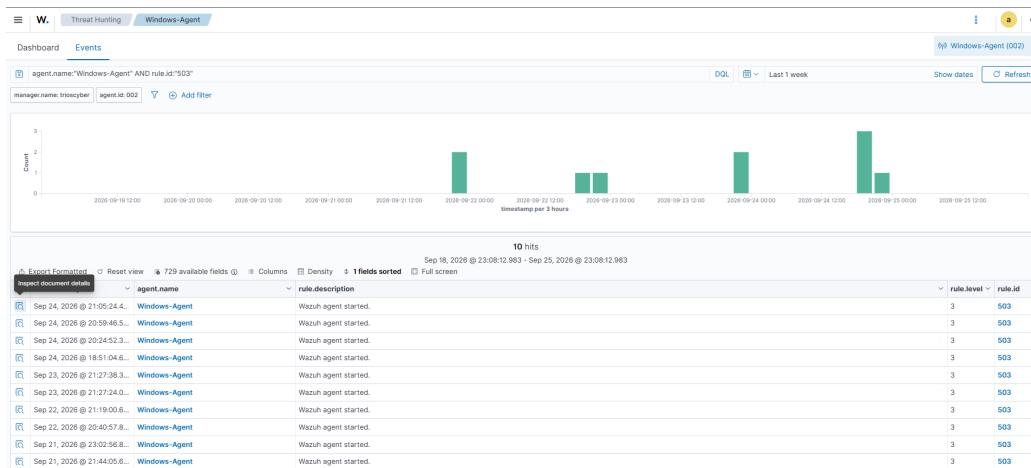
|                                 |                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rule.mitre.tactic               | Impact                                                                                                                                                                                                                                                                                                                                       |
| rule.mitre.technique            | Stored Data Manipulation                                                                                                                                                                                                                                                                                                                     |
| rule.nist_800_53                | SI.7                                                                                                                                                                                                                                                                                                                                         |
| rule.pci_dss                    | 11.5                                                                                                                                                                                                                                                                                                                                         |
| rule.tsc                        | PI1.4, PI1.5, CO6.1, CO6.8, CO7.2, CO7.3                                                                                                                                                                                                                                                                                                     |
| syscheck.attrs_after            | ARCHIVE                                                                                                                                                                                                                                                                                                                                      |
| syscheck.changed_attributes     | size, mtime, md5, sha1, sha256                                                                                                                                                                                                                                                                                                               |
| syscheck.event                  | modified                                                                                                                                                                                                                                                                                                                                     |
| syscheck.md5_after              | 4865668a965a5a0913a345cca5eacc40                                                                                                                                                                                                                                                                                                             |
| syscheck.md5_before             | 9fa301d3c1535d9926aafd7189316f02                                                                                                                                                                                                                                                                                                             |
| syscheck.mode                   | realtime                                                                                                                                                                                                                                                                                                                                     |
| syscheck.mtime_after            | Sep 24, 2026 @ 21:10:33.000                                                                                                                                                                                                                                                                                                                  |
| syscheck.mtime_before           | Sep 24, 2026 @ 21:07:34.000                                                                                                                                                                                                                                                                                                                  |
| syscheck.path                   | c:\users\matthew\documents\triosfimlab\day23_test.txt                                                                                                                                                                                                                                                                                        |
| syscheck.sha1_after             | 204d3d810ff670646158352bc1aeb2f87d7b2f3                                                                                                                                                                                                                                                                                                      |
| syscheck.sha1_before            | fdbcbfc305c74a7c94ef757a0b39bb208338c1d3                                                                                                                                                                                                                                                                                                     |
| syscheck.sha256_after           | ec915ee7f60e4754030cc50b78f9d1445d1bc45734c2857db35a80751d04968c                                                                                                                                                                                                                                                                             |
| syscheck.sha256_before          | b99ce48a98dd26181b55f592b87f31be3029181f2732581c791aba8a914d1d7                                                                                                                                                                                                                                                                              |
| syscheck.size_after             | 113                                                                                                                                                                                                                                                                                                                                          |
| syscheck.size_before            | 80                                                                                                                                                                                                                                                                                                                                           |
| syscheck.uid_after              | S-1-5-32-544                                                                                                                                                                                                                                                                                                                                 |
| syscheck.uname_after            | Administradores                                                                                                                                                                                                                                                                                                                              |
| syscheck.win_perm_after.allowed | DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, WRITE_ATTRIBUTES |
| syscheck.win_perm_after.name    | SYSTEM, Administradores, matthew                                                                                                                                                                                                                                                                                                             |
| timestamp                       | Sep 24, 2026 @ 21:10:36.667                                                                                                                                                                                                                                                                                                                  |

## 4. Alert 3 - Wazuh Agent Started

**Rule ID:** 503 **Alert level:** 3 **Agent:** Windows-Agent (ID 002, 10.10.20.30)  
**Description:** Wazuh agent started. **Timestamp:** Sep 24, 2026 @ 21:05:24.473

The filtered Threat Hunting results show multiple occurrences of the same agent-start event. The selected document identifies the source endpoint as Windows-Agent, location **wazuh-agent**, manager **trioscyber**, rule 503, and level 3. This type of event is useful for validating agent availability and for identifying unexpected service restarts or interruptions in endpoint monitoring.

**Recommended first triage action:** Confirm whether the agent start or restart was expected, then review preceding stop, disconnection, service-control, or communication events for evidence of an operational problem or unauthorized interruption.



### Document Details

Table JSON

```
{
  "t_index": "wazuh-alerts-4.x-2026.09.25",
  "t_agent.id": "002",
  "t_agent.ip": "10.10.20.30",
  "t_agent.name": "Windows-Agent",
  "t_data.extra_data": "Windows-Agent->any",
  "t_decoder.name": "ossec",
  "t_decoder.parent": "ossec",
  "t_full_log": "ossec: Agent started: 'Windows-Agent->any'.",
  "t_id": "1790301924.314907",
  "t_input.type": "log",
  "t_location": "wazuh-agent",
  "t_manager.name": "trioscyber",
  "t_rule.description": "Wazuh agent started.",
  "# rule.firedtimes": "1",
  "t_rule.gdpr": "IV_35.7.d",
  "t_rule.gpg13": "10.1",
  "t_rule.groups": "ossec",
  "t_rule.hipaa": "164.312.b",
  "t_rule.id": "503",
  "# rule.level": "3",
  "@ rule.mail": "false",
  "t_rule.nist_800_53": "AU.6, AU.14, AU.5",
  "t_rule.pci_dss": "10.6.1, 10.2.6"
}
```

```
{
  "t_rule.hipaa": "164.312.b",
  "t_rule.id": "503",
  "# rule.level": "3",
  "@ rule.mail": "false",
  "t_rule.nist_800_53": "AU.6, AU.14, AU.5",
  "t_rule.pci_dss": "10.6.1, 10.2.6",
  "t_rule.tsc": "CC7.2, CC7.3, CC6.8",
  "@ timestamp": "Sep 24, 2026 @ 21:05:24.473"
}
```

## 5. Conclusions

The Wazuh Rule and Alert Analysis was completed by examining three distinct events from Windows-Agent and documenting the required rule or alert level, description, agent, timestamp, and recommended first triage action. The selected records covered Windows service creation, file-integrity modification, and Wazuh agent startup. Together, they demonstrate how Wazuh event metadata can be used to establish context quickly and define an appropriate first response during SOC triage.